

# Electromagnetic Force from Temporal Asymmetry: U(1) Gauge Structure, DBI- $\Sigma$ Correspondence, and Modified Maxwell Equations

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The entropy-production parameter  $\Sigma = D(\rho_{\text{spacetime}} \parallel \rho_{\text{matter}})$  has been shown to unify gravitational phenomena from strong-field exponential metrics to weak-field galactic dynamics (Papers I–III). Here we prove that the U(1) gauge symmetry of electromagnetism is not an external input but a mathematical consequence of the  $\Sigma$  framework: the C\*-algebraic structure underlying quantum relative entropy, combined with the Poincaré representation constraint and the existence of tensor-product entanglement, uniquely selects complex quantum mechanics and hence U(1) as the minimal continuous gauge group (Theorem 1). We establish a DBI- $\Sigma$  correspondence (Theorem 2): the Born–Infeld Lagrangian of nonlinear electrodynamics and the ghost-condensation DBI action of the dark-matter Khronon share a universal  $\Sigma = -\ln \det$  structure inherited from D-brane geometry. We derive modified Maxwell equations in which gravitational entropy gradients couple to the electromagnetic field (Theorem 3), yielding a testable Coulomb correction  $\delta\Phi/\Phi \sim \beta r_s/r$  near compact objects, constrained by binary-pulsar data to  $|\beta| < 3 \times 10^{-3}$ .

## INTRODUCTION

Unifying gravity with the other forces remains the central open problem of fundamental physics. Einstein spent the last thirty years of his life pursuing a geometric unification of gravity and electromagnetism [4]—without success. Kaluza [5] and Klein [6] showed that a fifth compact dimension reproduces both Einstein’s equations and Maxwell’s equations from pure geometry, but at the price of an unexplained extra dimension and a massless dilaton. Weinberg [7] demonstrated that the long-range nature of electromagnetism requires U(1) gauge invariance, yet the *origin* of U(1) itself remained an axiom.

In the companion papers, we established the  $\Sigma$  framework: the Petz recovery fidelity bound  $F \geq e^{-\Sigma/2}$  [1] identifies the gravitational refractive index with information loss [2], and its weak-field limit reproduces galactic dark-matter phenomenology through Khronon condensation [3]. The central quantity is  $\Sigma = D(\rho_{\text{spacetime}} \parallel \rho_{\text{matter}})$ , the quantum relative entropy between the spacetime state and a matter reference state.

We now ask: does the  $\Sigma$  framework say anything about electromagnetism? We find three results:

(i) The mathematical structure that defines  $\Sigma$  (C\*-algebras, modular theory, tensor products, and the Poincaré group) uniquely selects complex quantum mechanics and hence U(1) as the minimal continuous gauge symmetry (Theorem 1) [NEW].

(ii) The Born–Infeld Lagrangian of nonlinear electrodynamics and the ghost-condensation DBI action of Paper III share a universal  $\Sigma = -\ln \det$  structure (Theorem 2) [NEW].

(iii) The Kaluza–Klein decomposition with a dynamical dilaton (amplitude of a complex Khronon) yields modified Maxwell equations in which  $\nabla_\nu \Sigma_{\text{grav}}$  appears

as a source (Theorem 3) [NEW].

We are explicit about what is derived, what is assumed, and what remains open. The framework does not predict the value of the fine-structure constant  $\alpha$ .

## WHY U(1) IS A CONSEQUENCE, NOT AN INPUT

The  $\Sigma$  framework rests on the quantum relative entropy  $D(\rho \parallel \sigma) = \text{Tr}(\rho \ln \rho - \rho \ln \sigma)$ , which presupposes a C\*-algebraic quantum theory on a Hilbert space. Over which number field? Three options exist: real ( $\mathbb{R}$ ), complex ( $\mathbb{C}$ ), and quaternionic ( $\mathbb{H}$ ), corresponding to the three associative normed division algebras by the Hurwitz theorem [11].

**Theorem 1** (U(1) from  $\Sigma$ ). *Within the  $\Sigma$  framework—defined by (a) a C\*-algebraic quantum theory, (b) Poincaré covariance, (c) tensor-product composability (required for entanglement and hence for  $\Sigma$  to be defined on composite systems), and (d) locality (existence of a causal net of algebras)—the scalar field underlying the Khronon is necessarily complex-valued, and its minimal continuous internal gauge symmetry is U(1).*

*Proof sketch.*—The argument proceeds in three steps.

*Step 1: Poincaré excludes  $\mathbb{R}$ .* Moretti and Oppio [8, 9] proved that for a quantum theory defined on a real Hilbert space, no continuous unitary representation of the Poincaré group exists that simultaneously satisfies the positive-energy condition and admits the standard spin-statistics connection. Specifically, the Wigner classification of irreducible representations requires a U(1) phase freedom to define half-integer spin; over  $\mathbb{R}$  this

fails. A subsequent experimental confirmation by Renou *et al.* [10] ruled out real quantum mechanics using a Bell-type inequality that is satisfied by all real Hilbert-space theories but violated by standard quantum mechanics—and by experiment.

*Step 2: Tensor products exclude  $\mathbb{H}$ .* Quaternionic Hilbert spaces do not admit a canonical tensor product [12, 13]: given two quaternionic Hilbert spaces  $H_1$  and  $H_2$ , the naive tensor product  $H_1 \otimes_{\mathbb{H}} H_2$  is not itself a quaternionic Hilbert space (the quaternionic inner product fails to be  $\mathbb{H}$ -linear on both sides). Since the  $\Sigma$  framework requires  $\Sigma = D(\rho_{AB} \parallel \rho_A \otimes \rho_B)$  for composite systems—and  $D$  is defined via  $\text{Tr}(\rho \ln \rho)$  on the tensor product—the absence of a tensor product in  $\mathbb{H}$  makes  $\Sigma$  undefined for composite systems [NEW]. This is not merely an inconvenience; it is a structural impossibility. No entanglement, no  $\Sigma$ , no framework.

*Step 3: Complex  $\Rightarrow U(1)$ .* Steps 1–2 select  $\mathbb{C}$ . A complex scalar field  $\phi$  has the global symmetry  $\phi \rightarrow e^{i\theta} \phi$  with  $\theta \in \mathbb{R}$ . The group of such transformations is  $U(1)$ . Promoting this to a local symmetry (required by the Doplicher–Roberts reconstruction theorem [14] for any superselection sector in the algebraic framework) introduces a  $U(1)$  gauge connection—the electromagnetic potential  $A_\mu$ .  $\square$

**Remark.** *Theorem 1 establishes that  $U(1)$  is the minimal continuous gauge symmetry forced by the  $\Sigma$  framework. It does not exclude larger gauge groups. In fact, the Szangolies hierarchy [19]—1-qubit  $\rightarrow U(1)$ , 2-qubit  $\rightarrow SU(2)$ , 3-qubit  $\rightarrow SU(3)$ —suggests that the full Standard Model gauge group  $SU(3) \times SU(2) \times U(1)$  may follow from multi-qubit entanglement structure. This remains a conjecture (Sec. ).*

**Remark.** *The Soler theorem [11] provides the mathematical backbone: any infinite-dimensional orthomodular lattice with the covering property is isomorphic to the lattice of closed subspaces of a Hilbert space over  $\mathbb{R}$ ,  $\mathbb{C}$ , or  $\mathbb{H}$ . Steps 1–2 then select  $\mathbb{C}$  uniquely. The experimental verification by Renou *et al.* [10]—published in *Nature*—confirms this selection empirically.*

## DBI- $\Sigma$ CORRESPONDENCE

### Born–Infeld electrodynamics

Born and Infeld [15] introduced a nonlinear modification of Maxwell’s theory with Lagrangian

$$\mathcal{L}_{\text{BI}} = b^2 \left( 1 - \sqrt{1 + \frac{F_{\mu\nu} F^{\mu\nu}}{2b^2} - \frac{(F_{\mu\nu} \tilde{F}^{\mu\nu})^2}{16b^4}} \right), \quad (1)$$

where  $b$  is a maximal field strength and  $\tilde{F}^{\mu\nu}$  is the Hodge dual. In modern language, this is the effective action on

a D-brane in string theory [16, 17]: the D-brane world-volume Lagrangian is  $\mathcal{L}_{\text{DBI}} \propto -\sqrt{-\det(g_{ab} + 2\pi\alpha' F_{ab})}$ . For purely electric fields in flat spacetime ( $\tilde{F} = 0$ ):

$$\mathcal{L}_{\text{BI}} = b^2 \left( 1 - \sqrt{1 - E^2/b^2} \right). \quad (2)$$

We define the Born–Infeld entropy production by analogy with the gravitational case:

$$\Sigma_{\text{EM}}^{\text{BI}} \equiv -\ln \left( 1 - \frac{E^2}{b^2} \right) = -\ln \det \left( \delta^a_b + \frac{F^a_b}{b} \right), \quad (3)$$

where the determinant form holds for the  $2 \times 2$  block  $(E, B)$ . This is manifestly non-negative, diverges as  $E \rightarrow b$  (the BI critical field), and vanishes for  $E = 0$ .

### Ghost-condensation DBI

In Paper III [3], the dark-matter sector was described by a Khronon field with the DBI kinetic function

$$K_{\text{DBI}} = \frac{2\mu^2}{\lambda_D} \left( 1 - \sqrt{1 - \lambda_D(Q - 1)^2} \right), \quad (4)$$

where  $Q = 1/N_\phi$  is the inverse Khronon lapse and  $\lambda_D$  is a dimensionless DBI parameter [18]. The Khronon entropy production is

$$\Sigma_{\text{grav}}^{\text{DBI}} \equiv -\ln(1 - \lambda_D \delta^2), \quad (5)$$

where  $\delta \equiv Q - 1$  is the perturbation away from the ghost condensate.

**Theorem 2** (DBI- $\Sigma$  correspondence). *The Born–Infeld electromagnetic entropy production  $\Sigma_{\text{EM}}^{\text{BI}}$  and the ghost-condensation DBI entropy production  $\Sigma_{\text{grav}}^{\text{DBI}}$  share a universal structure:*

$$\Sigma_{\text{DBI}} = -\ln \det(\mathbf{1} + \mathbf{X}), \quad (6)$$

where  $\mathbf{X}$  is the perturbation matrix:  $X^a_b = F^a_b/b$  for Born–Infeld, and  $X = \lambda_D \delta^2$  (scalar) for the Khronon. Both reduce to  $\Sigma \approx \text{Tr}(\mathbf{X})$  in the weak-field limit and diverge ( $\Sigma \rightarrow \infty$ ,  $\tau \rightarrow 1$ ) at the critical surface  $\det(\mathbf{1} + \mathbf{X}) = 0$ .

*Proof.*—The Born–Infeld Lagrangian on a  $Dp$ -brane in flat space is [16]

$$\mathcal{L} = -T_p \sqrt{-\det(\eta_{ab} + 2\pi\alpha' F_{ab})}. \quad (7)$$

Writing  $g_{ab} + 2\pi\alpha' F_{ab} = g_{ac}(\delta^c_b + g^{cd} \cdot 2\pi\alpha' F_{db})$ , we have

$$\det(\eta + F') = \det(\eta) \cdot \det(\mathbf{1} + \eta^{-1} F'), \quad (8)$$

so that the D-brane entropy production takes the form  $\Sigma = -\ln |\det(\mathbf{1} + \eta^{-1} F')|$ . For BI electrodynamics,  $\eta^{-1} F' \rightarrow F^a_b/b$ ; for the Khronon DBI, Eq. (4)

gives  $K_{\text{DBI}} = (2\mu^2/\lambda_D)(1 - \sqrt{1 - \lambda_D\delta^2})$ , and the corresponding entropy production is  $-\ln(1 - \lambda_D\delta^2)$ , which is  $-\ln\det(\mathbf{1} + \mathbf{X})$  with  $\mathbf{X} = -\lambda_D\delta^2$  (a  $1 \times 1$  matrix). Both cases yield Eq. (6).  $\square$

*Physical content.*—The correspondence is not accidental. Both the Born–Infeld action and the ghost-condensation DBI action descend from D-brane effective actions [16, 18]: the former from open-string modes on the brane (the electromagnetic field strength), the latter from the brane’s embedding coordinates (the Khronon scalar). The  $\Sigma = -\ln\det$  structure is the D-brane’s way of counting entropy: the determinant measures the “volume deformation” of the brane worldvolume induced by the field, and  $-\ln$  converts multiplicative deformation into additive entropy production [NEW].

The total entropy production decomposes as

$$\Sigma_{\text{total}} = \Sigma_{\text{grav}} + \Sigma_{\text{EM}} = 2\ln(Q_{\text{grav}} \cdot Q_{\text{EM}}), \quad (9)$$

where  $Q_{\text{grav}} = 1/\sqrt{-g_{00}}$  is the gravitational quality factor [2] and  $Q_{\text{EM}} \equiv 1/\sqrt{1 - E^2/b^2}$  is the electromagnetic analogue.

## MODIFIED MAXWELL EQUATIONS

### Complex Khronon and Kaluza–Klein decomposition

We now make the idea of a “complex Khronon” precise. Write the Khronon field as

$$\phi = f e^{i\theta}, \quad (10)$$

where the amplitude  $f > 0$  controls the gravitational sector ( $\Sigma_{\text{grav}} = 2\ln Q$  with  $Q$  determined by  $f$ ) and the phase  $\theta$  provides the U(1) degree of freedom identified in Theorem 1.

In a  $(4+1)$ -dimensional Kaluza–Klein framework, the five-dimensional metric decomposes as [5, 6]

$$ds_5^2 = e^{2\alpha\varphi} g_{\mu\nu} dx^\mu dx^\nu + e^{2\beta\varphi} (dy + A_\mu dx^\mu)^2, \quad (11)$$

where  $\varphi$  is the dilaton (related to  $f$  by  $e^{\beta\varphi} = f$ ),  $A_\mu$  is the electromagnetic potential (the “off-diagonal” metric component, identified with  $\partial_\mu\theta$  in the leading approximation), and  $\alpha, \beta$  are constants fixed by requiring canonical Einstein–Hilbert kinetic terms in four dimensions ( $\alpha = -1/(2\sqrt{3})$ ,  $\beta = \sqrt{3}\alpha$ ).

Standard Kaluza–Klein theory sets  $\varphi = \text{const}$  (frozen dilaton), recovering pure Maxwell theory. In the  $\Sigma$  framework the dilaton is dynamical—it *is* the gravitational sector  $\Sigma_{\text{grav}} = 2\ln Q$ .

**Theorem 3** (Modified Maxwell equations). *In the  $\Sigma$  framework with a dynamical dilaton  $\Sigma_{\text{grav}} = -\ln(-g_{00})$  and the Kaluza–Klein identification  $A_\mu \sim \partial_\mu\theta$ , the*

*electromagnetic field equations acquire a gravitational-entropy correction:*

$$\nabla_\nu F^{\mu\nu} + 2\beta_c (\nabla_\nu \Sigma_{\text{grav}}) F^{\mu\nu} + \mathcal{O}(F^3/b^2) = 4\pi J^\mu, \quad (12)$$

where  $\beta_c$  is a dimensionless coupling and the  $\mathcal{O}(F^3)$  term is the Born–Infeld nonlinear correction.

*Derivation.*—From the KK action with dynamical dilaton, the four-dimensional Maxwell sector picks up a dilaton coupling [21]:

$$S_{\text{EM}} = -\frac{1}{4} \int \sqrt{-g} e^{-2a\varphi} F_{\mu\nu} F^{\mu\nu} d^4x, \quad (13)$$

where  $a$  depends on the KK normalization. Varying with respect to  $A_\mu$ :

$$\nabla_\nu (e^{-2a\varphi} F^{\mu\nu}) = 4\pi J^\mu. \quad (14)$$

Expanding:

$$e^{-2a\varphi} [\nabla_\nu F^{\mu\nu} - 2a (\nabla_\nu \varphi) F^{\mu\nu}] = 4\pi J^\mu. \quad (15)$$

Identifying  $\Sigma_{\text{grav}} = 2\beta_0 \varphi$  (where  $\beta_0$  is the KK normalization constant), we obtain Eq. (12) with  $\beta_c = a/\beta_0$ . The BI nonlinear correction comes from replacing  $F_{\mu\nu} F^{\mu\nu}$  with the full BI determinant (1). In the limit  $\Sigma_{\text{grav}} = \text{const}$  (uniform gravity or flat space), Eq. (12) reduces to standard Maxwell. In the limit  $b \rightarrow \infty$  (weak fields), the BI correction vanishes.  $\square$

### Testable predictions

On a Schwarzschild background with  $\Sigma_{\text{grav}} = \ln(1/(1 - r_s/r))$  and  $\nabla_r \Sigma_{\text{grav}} = r_s/(r(r - r_s))$ , the modified Coulomb potential becomes

$$\Phi(r) = \frac{Q_e}{4\pi r} \left[ 1 + \beta_c \frac{r_s}{r} \ln\left(\frac{r}{r_s}\right) + \mathcal{O}(r_s^2/r^2) \right], \quad (16)$$

where  $Q_e$  is the electric charge and  $r_s = 2GM/c^2$  is the Schwarzschild radius. The correction is suppressed by  $r_s/r$  and therefore negligible at terrestrial scales ( $r_{s,\oplus}/R_\oplus \sim 10^{-9}$ ).

For compact objects the correction is significant. Near a neutron star with  $r_s/R \sim 0.4$ , the fractional Coulomb modification is  $\delta\Phi/\Phi \sim 0.4\beta_c \cdot \ln(R/r_s) \sim 0.4\beta_c$ .

Binary-pulsar timing constrains post-Keplerian parameters to  $\sim 10^{-3}$  precision [22, 23]. Since a modified Coulomb law would affect the binding energy of charged plasma in the magnetosphere, we estimate

$$|\beta_c| < 3 \times 10^{-3} \quad (\text{binary pulsar, 95\% CL}), \quad (17)$$

which is a derived constraint, not a tuned parameter [DERIVED].

**Remark.** Equation (12) with  $\beta_c = 0$  recovers standard Maxwell. The case  $\beta_c = 1$  corresponds to the simplest KK normalization. The binary-pulsar constraint eliminates the simplest KK model, requiring either  $\beta_c \ll 1$  (dilaton approximately frozen) or  $\beta_c = 0$  exactly (a symmetry protection). Whether the  $\Sigma$  framework predicts a specific value of  $\beta_c$  remains open.

## DISCUSSION

*Relation to Kaluza–Klein.*—The KK mechanism is a special case of our framework. Standard KK sets the dilaton to a constant, freezing  $\Sigma_{\text{grav}}$  and recovering pure Maxwell. The  $\Sigma$  framework dynamically relates the dilaton to  $\Sigma_{\text{grav}}$ : gravity and electromagnetism are the amplitude and phase of the same complex field, with the amplitude controlling information loss and the phase controlling charge. Unlike KK, the extra dimension here has a physical interpretation: it is the internal  $U(1)$  phase space mandated by complex quantum mechanics (Theorem 1), not an ad hoc geometric postulate.

*The Szangolies hierarchy.*—Szangolies [19] observed that single-qubit systems naturally carry  $U(1)$  structure (the Bloch sphere has a  $U(1)$  azimuthal symmetry), two-qubit entanglement introduces  $SU(2)$  (the entanglement group of the 2-qubit Hilbert space), and three-qubit systems generate the full  $SU(3)$  structure through the Cayley hyperdeterminant. If this hierarchy is physical:

**Conjecture** (Gauge group from entanglement). *The Standard Model gauge group  $SU(3) \times SU(2) \times U(1)$  is the entanglement symmetry group of the 3-qubit Hilbert space, arising naturally from the  $\Sigma$  framework applied to systems with  $n = 1, 2, 3$  entangled sectors.*

This is speculative. We record it because the  $\Sigma$  framework provides a concrete mechanism (Theorem 1) for the  $n = 1$  case that makes the general pattern non-trivial.

*DBI- $\Sigma$  as a conversion mechanism.*—The DBI- $\Sigma$  correspondence (Theorem 2) suggests a “rotation” between the gravitational and electromagnetic sectors: both contribute to  $\Sigma_{\text{total}} = \Sigma_{\text{grav}} + \Sigma_{\text{EM}}$ , and field configurations that maximize  $\Sigma_{\text{grav}}$  (strong gravity) can in principle convert into configurations that maximize  $\Sigma_{\text{EM}}$  (strong EM fields). This is reminiscent of the Gertsenshtein effect [20]—the conversion of gravitational waves to electromagnetic waves in a magnetic field—but the  $\Sigma$  framework suggests a more fundamental amplitude-phase mechanism underlying such conversions. The full nonlinear coupling between  $\Sigma_{\text{grav}}$  and  $\Sigma_{\text{EM}}$  is beyond the scope of this Letter.

*Fine-structure constant.*—The  $\Sigma$  framework determines the *structure* of the gauge group ( $U(1)$ ) but not the *coupling constant* ( $\alpha \approx 1/137$ ). In the Khronon language, the gravitational coupling is fixed by  $\mu_0 =$

$H_0/c$  [3], a cosmological input. Whether a similar cosmological boundary condition fixes  $\alpha$  is unknown. We note that the running of  $\alpha$  with energy scale admits a  $\Sigma = 2 \ln Q$  structure (with  $Q$  the electromagnetic quality factor), but the channel is independent of the gravitational one:  $\Sigma_{\text{EM}}$  and  $\Sigma_{\text{grav}}$  are produced by different quantum channels acting on different subsystems.

*What is derived, what is assumed, what is open.*—We summarize in Table I.

*Open problems.*—Three problems are immediate. First, the complete amplitude-phase coupling—the full nonlinear interaction between  $\Sigma_{\text{grav}}$  and  $\Sigma_{\text{EM}}$  beyond the linearized Eq. (12)—requires solving the coupled dilaton-Maxwell-Khronon system, which may modify the Gertsenshtein effect at the percent level near neutron stars. Second, the non-Abelian extension to  $SU(2)$  and  $SU(3)$  requires understanding multi-qubit entanglement structure within the  $\Sigma$  framework; the Szangolies hierarchy provides a road map but not a proof. Third, whether the binary-pulsar constraint  $|\beta_c| < 3 \times 10^{-3}$  saturates or admits further tightening from gravitational-wave observations (LIGO/Virgo binary mergers in strong-field regimes) is a concrete near-term test.

## CONCLUSIONS

We have shown that the  $\Sigma$  framework, originally developed for gravitational phenomena, extends naturally to electromagnetism.  $U(1)$  is not an input: it is forced by the mathematical requirements of the framework itself (complex Hilbert space from Poincaré covariance and tensor-product composability). The Born–Infeld and Khronon DBI actions share a universal  $\Sigma = -\ln \det$  structure inherited from D-brane geometry. Modified Maxwell equations with gravitational-entropy corrections are derived and constrained.

The emerging picture is one in which the complex

TABLE I. Epistemic status of results in this Letter.

| Result                                         | Status             |
|------------------------------------------------|--------------------|
| $\mathbb{R}$ excluded by Poincaré              | KNOWN [8]          |
| $\mathbb{H}$ excluded by tensor product        | KNOWN [12]         |
| $\mathbb{C} \Rightarrow U(1)$                  | KNOWN [14]         |
| Theorem 1 (synthesis)                          | NEW                |
| BI and DBI share $-\ln \det$                   | DERIVED            |
| Both from D-brane actions                      | KNOWN [16]         |
| Modified Maxwell (Thm. 3)                      | DERIVED            |
| $ \beta_c  < 3 \times 10^{-3}$                 | DERIVED (estimate) |
| $SU(3) \times SU(2) \times U(1)$ from $\Sigma$ | CONJECTURE         |
| Value of $\alpha$                              | OPEN               |

Khronon  $\phi = f e^{i\theta}$  encodes both gravity (through its amplitude) and electromagnetism (through its phase), with  $\Sigma = D(\rho_{\text{spacetime}} \parallel \rho_{\text{matter}})$  providing the unifying measure of departure from equilibrium. Whether the remaining gauge forces (weak, strong) follow from the same structure at higher entanglement order is the central open question.

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